

Q1. I am interested in doing an open source project in electronics, but the problem is that I am a computer science student with a very basic knowledge in electronics. I can also join a group to do a mini project or a major project in electronics. Are there any institutes that can help me with this?

T. Nitin Chander

Through email

A1. I, personally, don't think that you need a training institute to do an open source electronics project, being a computer science student yourself. Given below are some things you can do:

Arduino. Arduino is an open source electronics prototyping platform with flexible and easy-to-use software and hardware. It has a simple programming language specifically designed for non-programmers. This is a completely open source platform, for which you do not require much knowledge of electronics. A lot of community support and unlimited projects are available on the Internet.

Raspberry Pi. The Raspberry Pi is a credit-card-size single-board computer developed in the UK by the Raspberry Pi Foundation with the intention of promoting the teaching of basic computer science in schools. It has become a favourite playground for electronics enthusiasts. There is a library called 'Wiring Pi,' using which you can program Raspberry Pi in Arduino-like instructions. You can also program it in other programming languages like Python, C, PHP, etc.

Implementing a basic project would not require much electronics knowledge and there is great support from the community. Students with no engineering background from around the world are building amazing projects on these platforms.

EFY regularly publishes projects on both the platforms. You can start with a basic project and move ahead as you learn.

Q2. I am curious to know how to carry out the validation of

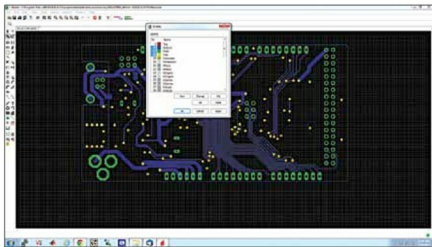


Fig. 1: Layers selection

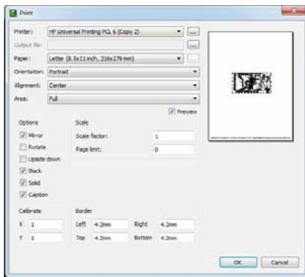


Fig. 2: Print options

manual soldering, and what are the parameters to be validated?

Nikita Usgaonkar

Through email

A2. Whether it is manual or automated, the standards remain the same. Please go through the following standards and you will have all the answers:

IPC-A-610	Acceptability of Electronic Assemblies
J-STD-001	Requirements for Soldered Electrical and Electronic Assemblies

The standards specifically cover all the aspects that need to be inspected and validated regarding electronic assemblies.

Q3. I am trying to print the bottom layer of my final layout in Eagle but when the design gets printed, it is much smaller than the original one. Please let me know how to correctly print the layout, and measure the actual size of the PCB to compare it with the printed file?

Kazi Yaser Arafat

Via email

A3. From your mail, it looks

like you have not correctly selected the scaling factor.

Please follow the steps indicated below:

1. Select the correct layers to be printed (shown in Fig. 1).
2. Select File→Print and select the options shown in Fig. 2.
3. If you are first making a PDF and then printing, take care that you print at 100 per cent and do not select 'Fit to the page.'

Answer compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/L, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender